ITA0605 - Machine Learning
List of Lab Exercises

17. To predict mobile prices using the Linear Regression algorithm.

Aim:

To predict mobile phone prices using the Linear Regression algorithm.

Algorithm:

1. Import the required Python libraries.
2. Load the mobile price dataset.
3. Separate the input features and target variable (mobile price).
4. Split the dataset into training and testing sets.
5. Train the Linear Regression model using the training data.
6. Predict mobile prices using the trained model.
7. Calculate the model performance using the R^2 score.
8. Display the predicted prices and model performance.

Dataset 1:

Program Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data = {
    'RAM': [4, 4, 6, 6, 8, 8, 12, 12, 16, 16],
    'Storage': [64, 128, 128, 256, 128, 256, 256, 512, 512, 1024],
    'Camera': [12, 16, 48, 50, 64, 64, 108, 108, 200, 200],
    'Price': [12000, 15000, 18000, 22000, 25000, 30000, 38000, 45000, 60000, 75000]
}

df = pd.DataFrame(data)

X = df[['RAM', 'Storage', 'Camera']]
y = df['Price']
model = LinearRegression()
model.fit(X, y)

new_sample = [[8, 256, 108]]
```

prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))

Output:

```
model.fit(X, y)

# New sample
new_sample = [[8, 256, 108]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

```
••• Predicted Mobile Price: ₹ 31317.27
```

Dataset 2:

Program Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
data = {
    'ProcessorSpeed': [1.8, 2.0, 2.2, 2.4, 2.6, 2.8, 3.0, 3.2, 3.4, 3.6],
    'RAM': [4, 4, 6, 6, 8, 8, 12, 12, 16, 16],
    'Battery': [4000, 4500, 5000, 5000, 5500, 6000, 6000, 6500, 7000, 7000],
    'Price': [10000, 12000, 17000, 21000, 26000, 32000, 40000, 50000, 62000, 76000]
}

df = pd.DataFrame(data)
X = df[['ProcessorSpeed', 'RAM', 'Battery']]
y = df['Price']
model = LinearRegression()
model.fit(X, y)
new_sample = [[2.9, 8, 6000]]
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

Output:

```
# New sample
new_sample = [[2.9, 8, 6000]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

```
••• Predicted Mobile Price: ₹ 34122.16
```

Dataset 3:

Program Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression


data = {
    'ScreenSize': [6.1, 6.2, 6.4, 6.5, 6.6, 6.7, 6.8, 6.9, 7.0, 7.1],
    'RefreshRate': [60, 90, 90, 120, 120, 120, 144, 144, 165, 165],
    'Battery': [4000, 4500, 5000, 5000, 5500, 6000, 6000, 6500, 7000, 7000],
    'Price': [12000, 15000, 18000, 23000, 28000, 34000, 42000, 52000, 65000, 78000]
}

df = pd.DataFrame(data)

X = df[['ScreenSize', 'RefreshRate', 'Battery']]
y = df['Price']

model = LinearRegression()
model.fit(X, y)

new_sample = [[6.7, 144, 6000]]

prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

Output:

```
model.fit(X, y)

# New sample
new_sample = [[6.7, 144, 6000]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))

••• Predicted Mobile Price: ₹ 42603.34
```

Dataset 4:

Program Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
data = {
    'RAM': [4, 6, 8, 8, 12, 12, 16, 16, 18, 24],
    'ProcessorCores': [6, 8, 8, 8, 8, 8, 8, 8, 8, 8],
    'Storage': [64, 128, 128, 256, 256, 512, 512, 1024, 1024, 1024],
    'Price': [14000, 19000, 26000, 32000, 42000, 52000, 65000, 82000, 95000, 120000]
}

df = pd.DataFrame(data)
X = df[['RAM', 'ProcessorCores', 'Storage']]
y = df['Price']

model = LinearRegression()
model.fit(X, y)

new_sample = [[12, 8, 512]]

prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

Output:

```
model.fit(X, y)

# New sample
new_sample = [[12, 8, 512]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

```
... Predicted Mobile Price: ₹ 53109.22
```

Dataset 5:

Program Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data = {
    'AntutuScore': [250000, 320000, 420000, 520000, 620000,
                    720000, 820000, 920000, 1020000, 1120000],
    'Battery': [4000, 4500, 5000, 5000, 5500,
                6000, 6000, 6500, 7000, 7000],
    'Camera': [48, 50, 64, 64, 108, 108, 200, 200, 200, 200],
    'Price': [15000, 18000, 24000, 30000, 38000,
              46000, 58000, 70000, 85000, 100000]
}

df = pd.DataFrame(data)

X = df[['AntutuScore', 'Battery', 'Camera']]
y = df['Price']
model = LinearRegression()
model.fit(X, y)
new_sample = [[750000, 6000, 108]]
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

Output:

```
model.fit(X, y)

# New sample
new_sample = [[750000, 6000, 108]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

```
••• Predicted Mobile Price: ₹ 55470.33
```

Result:

The Linear Regression algorithm was successfully implemented on all five mobile price datasets.
The model predicted the prices of the given new mobile samples based on their specifications.
Thus, mobile price prediction was successfully performed.